E-COMMERCE CUSTOMER INTELLIGENCE & REVENUE OPTIMIZATION

Business Analytics & Power BI Dashboard Report

Python • MySQL • Power BI • DAX

Portfolio Project | E-Commerce / Retail Analytics

Prepared for Data Analyst Portfolio & Interview Discussion

1. Executive Summary

This project converts e-commerce transaction data into a business-ready analytics solution covering sales performance, customer value, product/category performance, purchasing behavior, retention opportunities and operational patterns. Python was used for data preparation and customer analysis, MySQL for business-focused querying, and Power BI with DAX for interactive reporting.

The final Power BI report contains four analytical pages: Executive Overview, Product & Category, Customer Intelligence, and Behavior & Revenue Optimization. The dashboard is designed around common questions faced by e-commerce and retail teams rather than around technical charts alone.

TOTAL REVENUE	TRANSACTIONS / ORDERS	CUSTOMERS	AVERAGE ORDER VALUE
■2.18 Cr	**17,049**	**5,000**	**■1.3K**

AVERAGE CUSTOMER SPEND	REPEAT PURCHASE RATE	AT-RISK REVENUE	ANALYSIS PERIOD
■4.4K	**88.2%**	**■9.17 L** 4.2% of revenue	**2023-01 → 2024-03**

Portfolio value: The project demonstrates an end-to-end workflow from raw transactional data through cleaning, customer analytics, SQL analysis, DAX-based KPIs and interactive business reporting.

Important note

The figures in this report are based on the dashboard snapshot supplied for the project. Where a value is visually estimated from a chart rather than displayed as an exact KPI, it is described as approximate.

2. Business Problem & Objectives

Business Problem

Transactional data records what happened, but management needs to know why it matters and what action to take. The project therefore focuses on converting transactions into customer, product and revenue insights.

- Which categories contribute the most revenue and orders?
- How strong is repeat purchasing behavior?
- Which customer segments generate the highest revenue?
- Which customers may be at risk of becoming inactive?
- How are purchase frequency and customer spend related?
- Which cities, devices and payment methods contribute most to revenue?
- Does delivery time show a meaningful pattern with customer ratings?
- How does revenue change across the analysis period?

Project Objectives

- Clean and validate transactional data using Python.
- Build customer-level metrics and RFM segmentation.
- Prepare cohort/retention analysis for customer behavior.
- Use SQL to answer business questions with aggregations, rankings and reusable queries.
- Create an interactive Power BI report with clear KPI and cross-filtering behavior.
- Translate analytical results into practical e-commerce recommendations.

Target Users

E-commerce managers, category managers, customer/CRM teams, marketing teams and business stakeholders who need a concise view of sales performance and customer opportunities.

3. Data & Analytical Methodology

Dataset Profile

The project uses the E-Commerce Customer Behavior & Sales Analysis dataset. The working dataset contains transaction, customer, product, payment, device, delivery and rating attributes.

Field	Role in Analysis
order_id	Transaction / order identifier
customer_id	Customer-level analysis and RFM
date	Time trend and cohort preparation
product_category	Category performance
unit_price, quantity	Sales and basket metrics
discount_amount	Promotion / discount analysis
total_amount	Revenue metric
payment_method	Payment mix
device_type	Channel / device analysis
delivery_time_days	Operational experience
customer_rating	Customer experience indicator

Analytical Workflow

1	**Data validation**	Types, missing values, duplicates, date conversion
2	**Transformation**	Monthly fields and customer cohort preparation
3	**Customer analysis**	RFM scoring, segmentation and at-risk identification
4	**Retention**	Cohort month and months-since-first-purchase analysis
5	**SQL**	Business KPIs, category/customer analysis and ranking
6	**Power BI**	Model, DAX measures, slicers, visuals and navigation

4. Customer Intelligence & RFM Analysis

RFM analysis is used to move beyond transaction totals and understand customer value. Each customer is evaluated using three behavioral dimensions:

Metric	Meaning	Business Question
Recency	How recently the customer purchased	Who has not purchased recently?
Frequency	How often the customer purchased	Who purchases repeatedly?
Monetary	How much the customer spent	Who contributes the most revenue?

The dashboard groups customers into Champions, Loyal Customers, Potential Loyalists, At Risk and Lost Customers. This makes the analysis actionable: high-value active customers can be protected, while at-risk customers can be prioritized for re-engagement.

At-Risk revenue exposure shown in the dashboard is ■9.17 lakh, representing 4.2% of total revenue. This is a useful business signal because it quantifies the revenue currently associated with customers who may require retention attention.

Retention / Cohort Logic

- Convert transaction date into order month.
- Identify each customer's first purchase month as the cohort month.
- Calculate months since first purchase.
- Count unique customers by cohort and elapsed month.
- Use the resulting cohort structure to understand repeat activity over time.

5. KPI Snapshot & Business Interpretation

The dashboard's top KPI layer is intentionally compact so a business user can understand the overall situation before drilling into categories, customers or channels.

KPI	Dashboard value	Interpretation
Total Revenue	■2.18 Cr	Overall revenue generated in the selected reporting scope.
Total Orders	17,049	Transaction volume in the dashboard snapshot.
Total Customers	5,000	Distinct customers represented in the reporting scope.
Average Order Value	■1.3K	Average revenue per order.
Average Customer Spend	■4.4K	Average customer-level monetary contribution.
Repeat Purchase Rate	88.2%	High repeat activity in the displayed customer base.
At-Risk Revenue	■9.17 L / 4.2%	Revenue associated with the At-Risk customer segment.

Business Takeaways from the Dashboard Snapshot

- The customer base shows strong repeat-purchase behavior, with the dashboard reporting an 88.2% repeat purchase rate.
- Champions are the largest revenue-contributing RFM segment in the Customer Intelligence page, making high-value customer retention a priority.
- Electronics is the dominant revenue category in the displayed category analysis, followed by Home & Garden and Sports.
- Mobile contributes the highest device-level revenue in the Behavior & Revenue Optimization page.
- Credit Card is the largest payment-method revenue contributor in the displayed mix.
- The dashboard exposes a measurable At-Risk revenue pool, allowing retention activity to be prioritized by financial impact.

These findings should be treated as decision-support signals. Before using them for live business decisions, a production team would validate them against current-period data, campaign history, margins and operational targets.

6. Power BI Dashboard Architecture

The report is structured as a four-page analytical journey. Global filters sit at the top of each page, allowing users to change the month range, category, customer segment, city, payment method, device and returning-customer status.

Page	Purpose	Primary Questions
Executive Overview	Management summary	What is happening to revenue, orders and customer activity?
Product & Category	Category performance	Which categories generate revenue and how do discounts relate to sales?
Customer Intelligence	Customer value & risk	Who are the most valuable and at-risk customers?
Behavior & Revenue Optimization	Channel & experience	Which devices, cities and payment methods contribute revenue? How does

- Cross-filtering allows a user to click a category, segment or chart element and inspect its effect elsewhere.
- The navigation bar separates executive, product, customer and behavior questions without overloading one page.
- The KPI row maintains consistent business measures across pages for easier comparison.
- The design uses restrained colors, clear hierarchy and compact visual density suitable for a business reporting context.

7.1 Dashboard Page — Executive Overview

Management-level view of revenue trend, category contribution and new versus returning customer activity.

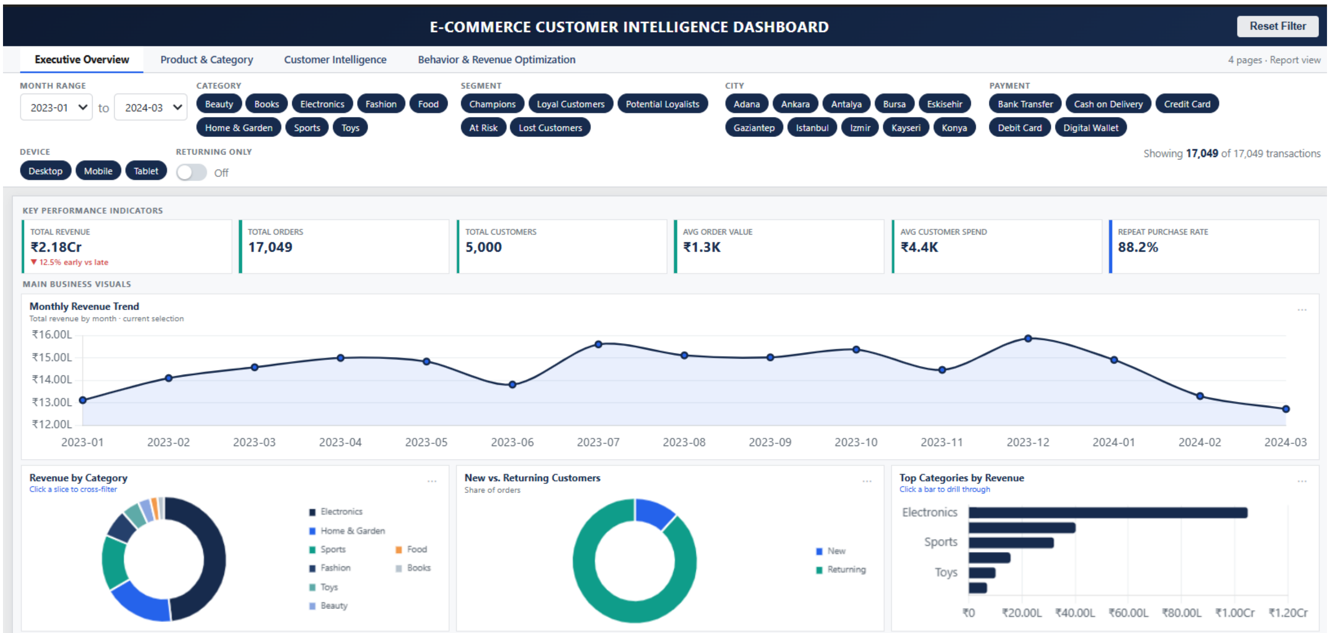

Figure 1: Supplied Power BI dashboard screenshot — Executive Overview

What this page communicates

- Monthly revenue is shown across Jan 2023 to Mar 2024, with a visible peak around Dec 2023 in the supplied snapshot.
- Electronics is the largest revenue category in the displayed overview.
- The New vs Returning visual shows returning customers dominating the order mix in the snapshot.

The Executive Overview is intentionally designed as the first page a recruiter, manager or stakeholder can open to understand the project in seconds.

7.2 Dashboard Page — Product & Category

Category performance view combining revenue ranking, discount-versus-revenue analysis and a detailed category matrix.

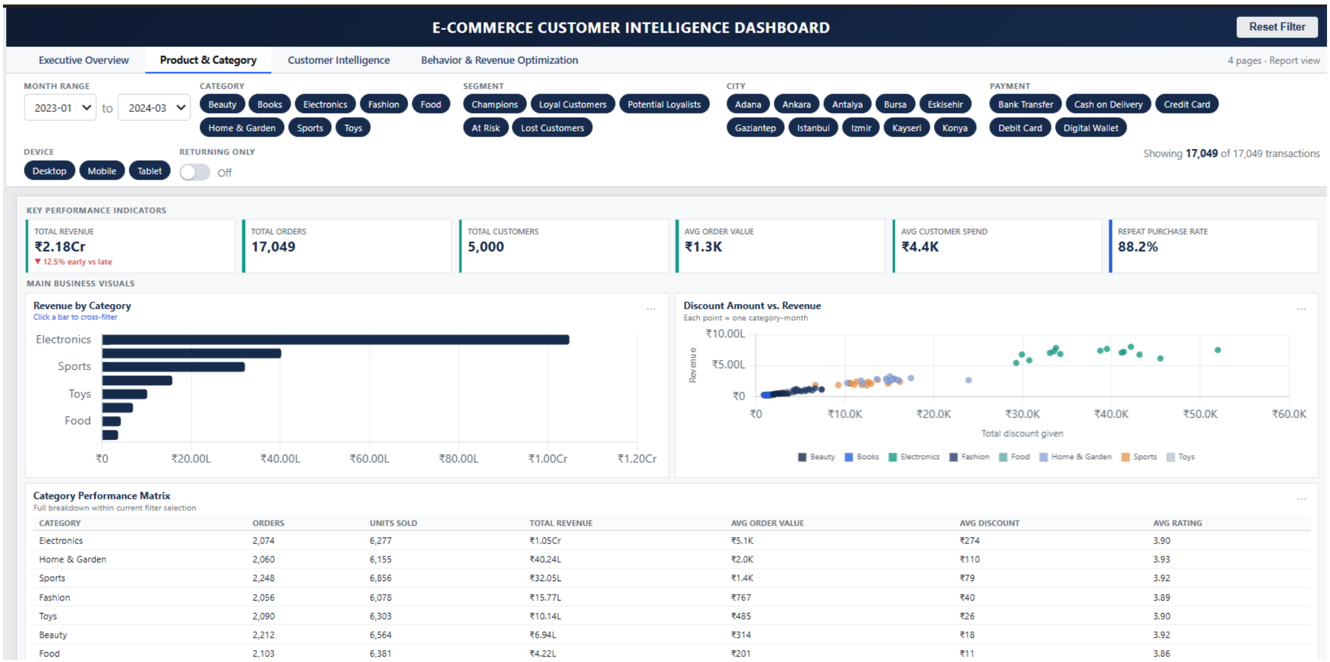

Figure 2: Supplied Power BI dashboard screenshot — Product & Category

What this page communicates

- Electronics contributes approximately ■1.05 Cr in the displayed matrix, followed by Home & Garden at approximately ■40.24 L and Sports at approximately ■32.05 L.
- The matrix combines orders, units sold, revenue, average order value, average discount and average rating.
- The discount-versus-revenue scatter helps identify category-month combinations with higher discount investment and corresponding revenue.

7.3 Dashboard Page — Customer Intelligence

Customer value and risk view centered on RFM segmentation and revenue exposure.

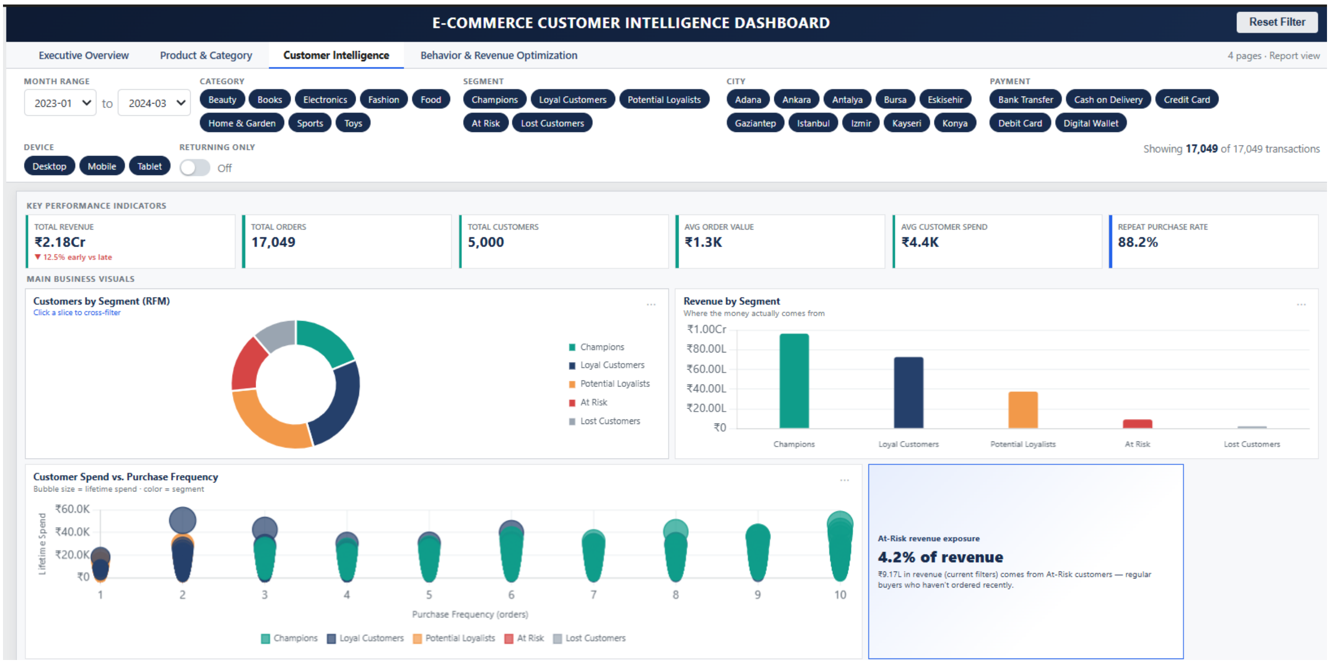

Figure 3: Supplied Power BI dashboard screenshot — Customer Intelligence

What this page communicates

- Champions and Loyal Customers account for the strongest revenue contribution in the displayed segment chart.
- The dashboard identifies an At-Risk revenue exposure of ■9.17 L, equal to 4.2% of displayed revenue.
- The spend-versus-frequency scatter supports customer-level interpretation by showing how lifetime spend changes with purchase frequency.

This is the strongest differentiating page for the portfolio because it moves the project beyond basic sales reporting into customer value and retention analysis.

7.4 Dashboard Page — Behavior & Revenue Optimization

Channel, geography, payment and customer-experience view.

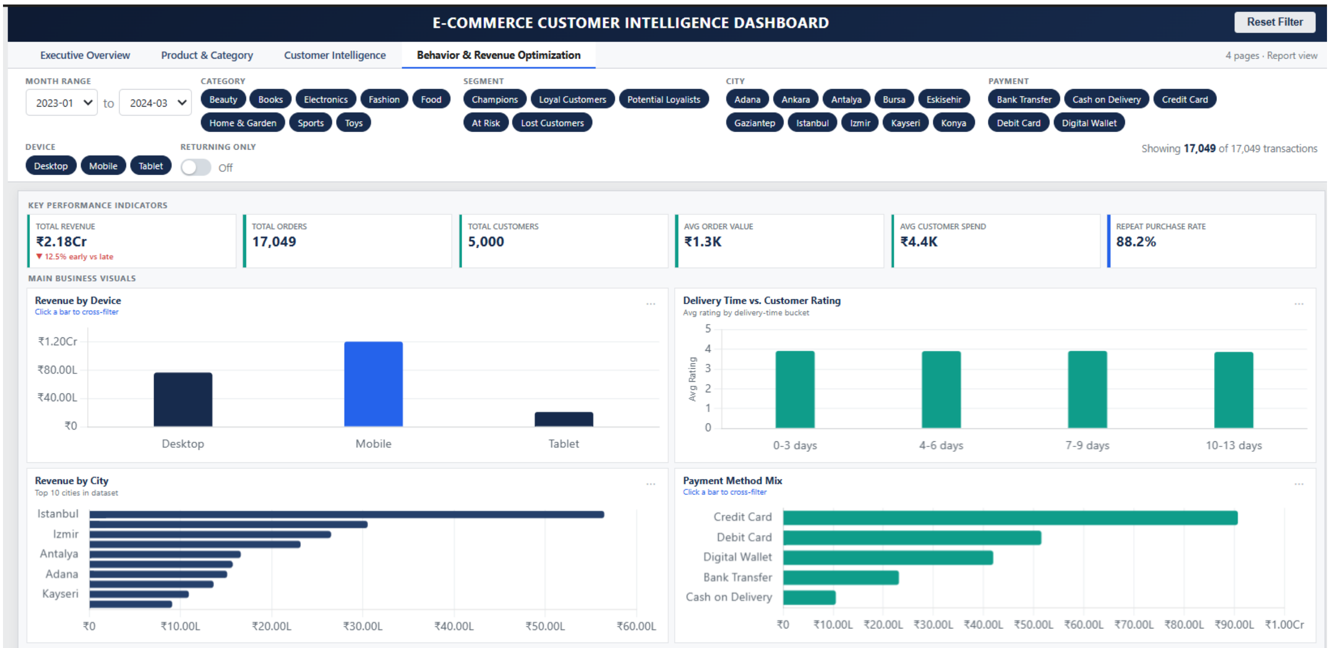

Figure 4: Supplied Power BI dashboard screenshot — Behavior & Revenue Optimization

What this page communicates

- Mobile is the largest device-level revenue contributor in the displayed chart.
- Istanbul is the leading city by revenue in the displayed Top 10 city view.
- Credit Card is the largest payment-method revenue contributor in the displayed mix.
- Average customer ratings remain close to 3.9 across the delivery-time buckets shown, so the visual is better used for monitoring than for claiming a strong causal relationship.

8. Business Recommendations

The dashboard is designed to support action. The following recommendations connect the observed analytical patterns to realistic e-commerce decisions.

Prioritize At-Risk Customers

Use RFM-based At-Risk identification to create re-engagement audiences. Prioritize customers with higher monetary value first.

Protect High-Value Customers

Champions and Loyal Customers should receive loyalty benefits, personalized recommendations and proactive retention treatment.

Optimize Category Investment

Electronics is the dominant revenue category in the snapshot. Management can protect availability and customer experience while reviewing the economics of promotional spend.

Review Discount Efficiency

Use the discount-versus-revenue view to distinguish productive promotions from discounting that does not translate into proportionate revenue.

Strengthen Mobile Experience

Mobile is the strongest device-level revenue contributor in the snapshot. Checkout, navigation and payment experience should therefore be monitored closely on mobile.

Monitor Delivery Experience

Track delivery-time buckets together with customer ratings. The displayed snapshot does not show a large rating gap, so operational changes should be validated with longer-term data.

Use City-Level Prioritization

Use the city revenue ranking to guide inventory, campaigns and service-level planning, while validating demand and margin before reallocating resources.

9. Technical Implementation

Python

- Pandas and NumPy for data inspection, transformation and aggregation.
- Datetime conversion using the dataset's actual date field.
- Customer-level aggregation for Recency, Frequency and Monetary analysis.
- RFM segment creation and At-Risk customer extraction.
- Cohort month and months-since-first-purchase preparation for retention analysis.

MySQL

- Revenue and order KPIs using aggregations.
- Category and customer performance analysis.
- Ranking and top-N analysis.
- CTEs and window functions for reusable business logic.
- Date-based monthly reporting.

Power BI & DAX

- Four-page report navigation.
- KPI cards for executive measures.
- Interactive slicers for time, category, segment, city, payment, device and returning-customer status.
- Cross-filtering between business visuals.
- RFM segment visuals and revenue contribution.
- Category performance matrix and operational behavior visuals.

Recommended GitHub Deliverables

- Cleaned dataset or a documented sample, subject to dataset licensing and file-size limits.
- Python notebook with clear section headings and outputs.
- MySQL SQL script containing the main business queries.
- Power BI .pbix file where sharing is appropriate.
- Four dashboard screenshots.
- This business report PDF.
- README.md explaining the business problem, workflow, tools, insights and recommendations.

10. Conclusion

The E-Commerce Customer Intelligence & Revenue Optimization project demonstrates a complete fresher-level analytics workflow with a strong e-commerce/retail focus. Rather than stopping at descriptive sales charts, the solution connects revenue performance with customer value, RFM segmentation, retention thinking, category performance and operational behavior.

The Power BI dashboard provides a clear stakeholder interface, while Python and SQL provide the analytical foundation behind it. Together, these components demonstrate practical skills expected from an entry-level Data Analyst working with transactional and customer data.

The next natural extensions are customer lifetime value, churn prediction, sales forecasting, campaign effectiveness and more detailed cohort retention. These should be added only when the underlying business question and data support them.

Portfolio positioning: E-Commerce / Retail Analytics • Customer Analytics • Revenue Analytics • Business Intelligence

End of Report